CAMBRIDGE
INTERNATIONAL EXAMINATIONS

NOVEMBER 2002

INTERNATIONAL GCSE

MARK SCHEME

MAXIMUM MARK : 80

SYLLABUS/COMPONENT : 0625/2

PHYSICS
(CORE)

UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

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QU.	SCHEME	TARGET GRADE	MARK
1. (a) (i)	greater	F	M1
(ii)	P.E. (or equiv.) has increased OR work done lifting case	F	A1
(b) (i)	greater	F	M1
(ii)	it is moving OR now has K.E. (or equiv.)	F	A1
			4
2. (a)	insulator	F	B1
(b)	radiation	F	B1
(c)	conductor	F	B1
(d)	convection	F	B1
			4
3. (a)	arrow(s) clockwise	C	B1
(b)	3 circles (by eye) around wire (need not be concentric, ignore other lines)	F	B1
	circles concentric with wire (by eye)	C	B1
			3
4. (a) (i)	1020 - 610	F	C1
	410 (g)	F	A1
(ii)	mass/volume	F	C1
	his (i)/500	F	C1
	0.82 e.c.f.	F	A1
	g/cm^3	C	B1
(iii)	use measuring cylinder/pipette/narrower jug /burette	C	B1
(b)	level shown below oil level	C	B1
			8

5.	(a)	changes into a different nucleus/substance/isotope/nuclide OR loses/emits part of itself/particles OR loses/emits an alpha/beta particle/gamma ray OR mass decreases OR different mass no.	F	B1
	(b)	evidence of 2 half-lives	C	C1
		56 (years)	C	A1 3
6.	(a)	temperature NOTHING ELSE	F	B1
		solid turns to liquid OR liquid turns to solid	F	B1
	(b)	last 2 both ticked	C	B1
	(c)	(i) horizontal straight line (nothing else)	F	B1
		(ii) B.P. correctly marked at horizontal line (condone extras) allow 100°C i.e. on temp axis MUST BE CLEAR	C	B1 5
7.	(a)	rub them together	F	B1
	(b)	G.L.E. (leaf deflects) OR pick up fluff etc OR crackles when discharged OR makes hair rise etc	F	B1
	(c)	region (or equiv.)	C	B1
		where electric charge experiences a force/attraction/repulsion NOT "effect"	C	B1
	(d)	(i) moves away/repel/deflects/spins	F	M1
		(ii) like charges (NOT poles) repel	F	A1
	(e)	copper is a conductor (or similar comment)/copper can't be charged B0 for conductor of heat	C	B1 7
8.	(a)	volt OR volts OR V	F	B1
	(b)	resistance = p.d./current in any form, allow symbols or mixture (allow B1 for just p.d./ current)	2F	B2
	(c)	4.7 = V/0.5	F	C1
		2.35 (V)	F	A1
	(d)	(i) increases OR is a maximum	F	B1
		(ii) decreases OR minimum condone "to zero"	F	B1
	(e)	10 - 4.7	C	C1
		5.3 (Ω)	C	A1 9

9.	(a)	avoid problems with echoes	C	B1
	(b)	time would have been too small to measure (with stopwatch) OR to give a greater time interval OR for accuracy	C	B1
	(c)	tape-measure OR trundle wheel OR metre rule OR range-finder OR calibrated strides	F	B1
	(d)	light travels fast/ instantaneously/ at 3×10^8 m/s	C	B1
		sound travels slowly/ slower/ at 330 (±30) m/s	F	B1
		(Note: "sound travels much slower than light" OR "light travels much faster than sound" scores B1,B1) "sound travels slower than light" etc gets B1, B0		
	(e)	speed = distance/time allow $s = 2d/t$	F	C1
		238/0.7	F	C1
		340	F	A1
		m/s	C	B1
	(f)	effect of air movement OR take average OR repetition to check NOT "for accuracy", unless adequately explained	C	B1 10

10.	(a)	(i)	moves (ignore any direction) NOT vibrates	F	B1
		(ii)	conductor experiences force in magnetic field	C	B1
			current-carrying conductor	C	B1
		(iii)	moves in opposite direction to (i)	F	B1
	(b)	(i)	commutator OR split ring allow commuter NOT slip rings	C	B1
			brush OR contact NOT spring	F	B1
			magnet OR pole	F	B1
		(ii)	commutator OR split ring e.c.f. from (i)	C	B1
		(iii)	rotates / moves other way / to the left rotates anticlockwise	F	B1 9

allow answers in form of current in field experiences a force

11. (a)	current causes magnetic field	F	B1
	iron reeds magnetised	C	B1
	magnetised in same direction OR adjacent ends opposite polarity	C	B1
	(ends) attract each other	C	B1
(b)	temperature rises	F	B1
	resistance decreases	F	B1
	(eventually) enough current to close relay	C	B1
	current flows in lamp circuit or equiv.	C	B1
			8

12. (a) (i)	ray refracted down at A } condone dispersion; Mark worst ray	F	M1
	not below normal	C	A1
	refracted down at 2nd surface	F	B1
(ii)	refraction / refracted OR deviation	F	B1
(b)	violet greater refraction than red ~~at A~~	F	B1
	2 rays diverging all the way to the screen from A condone repetition of errors in (i)	C	B1
(c)	spectrum (or equiv.) OR colours OR rainbow NOT dispersion	F	B1
(d) (i)	X marked ~~above~~ beyond position of red	F	B1
(ii)	not in visible spectrum OR invisible	C	B1
(iii)	any example of a suitable I.R. detector NOT "IR/heat sensor/detector"	C	B1
			10